

Gini Gain

Gini impurity gives us a measure of the impurity of a tree at given depth. The best split is the one that produces the largest decrease (or *Gini gain*) in the resulting child nodes.

In our example tree, the Gini impurity of

- ▶ The Gini impurity of the:
 - ▶ starting node = $1 - (\frac{1}{3}^2 + \frac{1}{3}^2 + \frac{1}{3}^2) = \frac{2}{3} = 0.666$
 - ▶ left child node = $1 - (0.87^2 + 0.12^2 + 0.02^2) = 0.23$
 - ▶ right child node = $1 - (0.05^2 + 0.45^2 + 0.50^2) = 0.62$
- ▶ So the weighted average of the two child nodes was $0.35 * 0.23 + 0.65 * 0.62 = 0.434$, since
 - ▶ 35% of the original 150 irises are in the left child node
 - ▶ 65% of the original 150 irises are in the right child node
- ▶ So the Gini gain is $0.666 - 0.434 = 0.233$. i.e. the Gini impurity decreased by 0.233 closer towards 0 (perfect purity)